

UICT — Uganda
Department of Computer Science

BSM 121

Research Methods

Academic Research, Methodology & Report Writing

Ground to Mastery Study Notes

Bravehart
DCS Year 1, Semester 2
Academic Year 2025/2026

Assessment: Coursework 40% | Final Exam 60%

Assessment Breakdown

Component	Weight	Details
Continuous Assessment (Coursework)	40%	Tests, assignments, practicals
Final Examination	60%	Written final exam at semester end

Minimum pass mark: 50% overall

You must pass BOTH components to pass the course unit.

Unit 1: Introduction to Research

Research is a systematic, disciplined process of gathering, analysing, and interpreting information to answer questions or solve problems. It is the foundation of knowledge creation in all academic and professional fields.

Term	Definition
Research	A systematic investigation to establish facts, test hypotheses, or develop new knowledge
Scientific method	A standardised process for conducting research: observe, hypothesise, experiment, analyse, conclude
Hypothesis	A tentative, testable statement predicting the relationship between variables
Theory	A well-tested explanation of observed phenomena, supported by extensive evidence
Variable	Any characteristic, attribute, or factor that can change or take different values
Objectivity	Conducting research without personal bias influencing the process or findings
Reliability	The consistency of a research instrument — repeated measurements yield the same results
Validity	Whether a research instrument actually measures what it is intended to measure

Characteristics of Good Research

- Empirical — based on direct or indirect observation and experience
- Systematic — follows an orderly, structured process
- Logical — uses rational reasoning and sound arguments
- Replicable — other researchers can repeat the study and get similar results
- Objective — findings are free from personal bias
- Ethical — respects participants and follows established ethical guidelines

EXAM TIP: Know the difference between basic research (pure; aimed at expanding knowledge without immediate practical application) and applied research (aimed at solving specific real-world problems). ICT research is typically applied research.

Unit 2: Types of Research

Research can be classified according to its purpose, approach to data collection, and the nature of the data collected. Understanding these distinctions helps in selecting the appropriate research design for a study.

Research Type	Description	Example
Qualitative Research	Explores experiences, meanings, and perspectives; uses words and descriptions	Interviewing ICT students about e-learning challenges
Quantitative Research	Measures variables numerically; uses statistics and mathematical analysis	Surveying 200 students on internet speed satisfaction
Descriptive Research	Describes the characteristics of a phenomenon as it currently exists	Survey on smartphone usage patterns among UICT students
Exploratory Research	Investigates a new or poorly understood problem	Exploring blockchain adoption in Ugandan banks
Explanatory (Causal) Research	Examines cause-and-effect relationships between variables	Effect of internet access on student academic performance
Action Research	Practical problem-solving research within a specific context by practitioners	A teacher researching how to improve ICT lesson delivery

EXAM TIP: A mixed-methods approach combines qualitative and quantitative research. It gives a more complete picture — numbers provide breadth while words provide depth. Know when to use each: use quantitative for generalisation, qualitative for understanding.

Unit 3: The Research Process

The research process is a systematic series of steps followed to conduct a study from identifying a problem to communicating findings. Each step builds upon the previous one.

Steps in the Research Process

- 1. Identify and define the research problem — what do you want to investigate?
- 2. Review existing literature — what is already known about the topic?
- 3. Formulate research objectives and questions — what specific answers are you seeking?
- 4. State the hypothesis (if applicable) — what is your prediction?
- 5. Design the research methodology — how will you collect and analyse data?
- 6. Collect data — implement the chosen data collection methods
- 7. Analyse and interpret data — apply statistical or qualitative analysis
- 8. Draw conclusions — what do the findings mean?
- 9. Write and present the research report — communicate findings formally
- 10. Suggest recommendations — what actions should be taken based on findings?

Term	Definition
Research problem	A gap in knowledge or an unresolved issue that the research seeks to address
Research objective	A specific, measurable statement of what the study aims to achieve
Research question	A specific question that the study is designed to answer
Scope	The boundaries and limitations of what the research covers
Limitation	Factors beyond the researcher's control that may affect research results

EXAM TIP: Research objectives must be SMART: Specific, Measurable, Achievable, Realistic, and Time-bound. Objectives should start with action verbs: 'To examine...', 'To determine...', 'To assess...', 'To evaluate...'.

Unit 4: Literature Review

A literature review is a critical analysis and synthesis of existing published research related to a study topic. It demonstrates the researcher's knowledge of the field and identifies gaps that the current study will address.

Purposes of a Literature Review

- Establishes what is already known about the topic
- Identifies gaps, contradictions, or unresolved questions in existing research
- Provides theoretical and conceptual frameworks for the study
- Justifies the need for the current research
- Helps refine research questions and hypotheses
- Avoids duplication of existing work

Sources for Literature Review

- Journal articles (peer-reviewed) — most credible academic sources
- Books and textbooks — foundational knowledge
- Conference papers — cutting-edge, recent findings
- Theses and dissertations — detailed academic research
- Government reports and official statistics — authoritative data
- Online databases: Google Scholar, IEEE Xplore, ACM Digital Library, JSTOR

Term	Definition
Primary source	Original research or firsthand accounts (journal articles, theses, raw data)
Secondary source	Analysis or interpretation of primary sources (textbooks, review articles)
Citation	A reference to a source used in a research paper
APA format	American Psychological Association referencing style widely used in ICT research
Plagiarism	Using someone else's work or ideas without proper attribution — an academic offence

EXAM TIP: A good literature review is thematic, not just a list of summaries. Group sources by themes or arguments. Use signal phrases: 'According to...', 'Smith (2020) argues that...', 'Studies by X and Y suggest...'. Always cite every source you use.

Unit 5: Research Design & Methodology

Research design is the blueprint for the entire research study. It specifies the methods and procedures for collecting, measuring, and analysing data to achieve the research objectives.

Term	Definition
Research design	The overall strategy and plan for conducting the research study
Research philosophy	The beliefs and assumptions underlying how knowledge is gathered (positivism, interpretivism)
Population	The entire group of individuals that the research is concerned with
Sample	A selected subset of the population chosen to represent the whole group
Sampling frame	A complete list of all individuals in the target population from which a sample is drawn
Sample size	The number of participants included in the study
Pilot study	A small-scale preliminary study to test research instruments before the main study

Sampling Methods

- Random sampling — every member of the population has an equal chance of selection (eliminates bias)
- Stratified sampling — population is divided into subgroups (strata); samples taken from each
- Systematic sampling — every nth person on the list is selected
- Cluster sampling — population is divided into clusters; whole clusters are randomly selected
- Convenience sampling — most accessible participants are selected (less rigorous but practical)
- Purposive sampling — participants selected deliberately based on specific characteristics

EXAM TIP: Random sampling gives the highest validity for generalisation. Purposive sampling is best for qualitative research where specific expertise or experience is needed. Convenience sampling is common in student projects but has the lowest generalisability.

Unit 6: Data Collection Methods

Data collection is the process of gathering information from sources to answer research questions. The choice of method depends on the research design, type of data needed, resources, and access to participants.

Method	Description	Best For
Questionnaire/Survey	Structured written questions administered to participants	Quantitative data from large groups
Interview	Verbal exchange between researcher and participant	In-depth qualitative insights
Observation	Researcher directly watches and records behaviour or events	Studying processes or behaviour in context
Focus Group	Group discussion guided by a moderator	Exploring group opinions and shared experiences
Document Analysis	Examining existing documents, reports, or records	Historical data or policy analysis
Experiment	Manipulating variables to measure effects	Establishing cause-and-effect relationships

Types of Questionnaire Questions

- Closed questions — fixed options (Yes/No, multiple choice, Likert scale); easy to analyse
- Open questions — free-form written answers; rich qualitative data; harder to analyse
- Likert scale — 5 or 7-point scale from Strongly Disagree to Strongly Agree; measures attitudes
- Ranking questions — respondent orders options by preference

EXAM TIP: Primary data is collected directly by the researcher for the specific study (interviews, surveys, experiments). Secondary data already exists and was collected for another purpose (books, reports, databases). Both can be used in the same study.

Unit 7: Data Analysis & Interpretation

Data analysis transforms raw data into meaningful findings. The method of analysis depends on whether the data is quantitative (numerical) or qualitative (descriptive).

Quantitative Data Analysis

- Descriptive statistics — summarise data: mean (average), median (middle value), mode (most frequent), standard deviation (spread)
- Frequency distribution — shows how often each value occurs
- Percentage and proportion — express parts of the whole
- Charts and graphs — bar charts, pie charts, histograms, line graphs visualise data
- Correlation — measures the strength and direction of the relationship between two variables
- Inferential statistics — draw conclusions about a population from sample data (chi-square, t-test)

Qualitative Data Analysis

- Transcription — converting audio/video interviews to text
- Coding — labelling segments of text with categories or themes
- Thematic analysis — identifying recurring themes and patterns in qualitative data
- Content analysis — systematically counting or categorising communication content
- Narrative analysis — interpreting the stories and experiences shared by participants

EXAM TIP: Know the measures of central tendency: Mean = sum/count (affected by outliers); Median = middle value (not affected by outliers); Mode = most frequent value. For skewed data, median is a better measure than mean.

Unit 8: Research Ethics & Report Writing

Research ethics ensures that studies are conducted with integrity and respect for participants. A well-structured research report communicates findings clearly and professionally following academic conventions.

Key Ethical Principles in Research

- Informed consent — participants must be told about the study and agree voluntarily
- Confidentiality — personal information shared by participants must not be disclosed
- Anonymity — participants' identities should not be revealed in the report
- Do no harm — the study must not physically, psychologically, or socially harm participants
- Voluntary participation — participants can withdraw at any time without penalty
- Honesty and integrity — data must not be fabricated, falsified, or selectively reported
- Plagiarism avoidance — all sources must be properly cited and credited

Structure of a Research Report

- Title Page — title, author, institution, date
- Abstract — brief summary (150–250 words) of objectives, methodology, findings, conclusion
- Table of Contents
- Chapter 1: Introduction — background, problem statement, objectives, significance, scope
- Chapter 2: Literature Review — theoretical framework, review of related studies
- Chapter 3: Methodology — research design, population, sampling, data collection, analysis methods
- Chapter 4: Data Presentation & Analysis — findings presented with tables/charts
- Chapter 5: Discussion, Conclusions & Recommendations
- References/Bibliography
- Appendices — questionnaires, raw data, consent forms

EXAM TIP: The abstract must stand alone — a reader should understand the entire study from the abstract alone. It includes: purpose, methodology, key findings, and conclusions. Do NOT include citations in the abstract.